

Machine Learning: General Concepts

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 **anr**®
agence nationale
de la recherche

What we will cover on this talk?

- Artificial Intelligence & Machine Learning
- How do we learn: Training and Supervision
- Vocabulary: Clustering, Classification, Regression
 - Artificial Neural Networks
- Preparing for hands-on

Please, help me by making questions!

This is a very specific topic, potentially new for many here. If not now, be free to stop me at some moment on the conference and ask!

"We all know something. We all choose to ignore something. And that's why we always learn." (Paulo Freire)

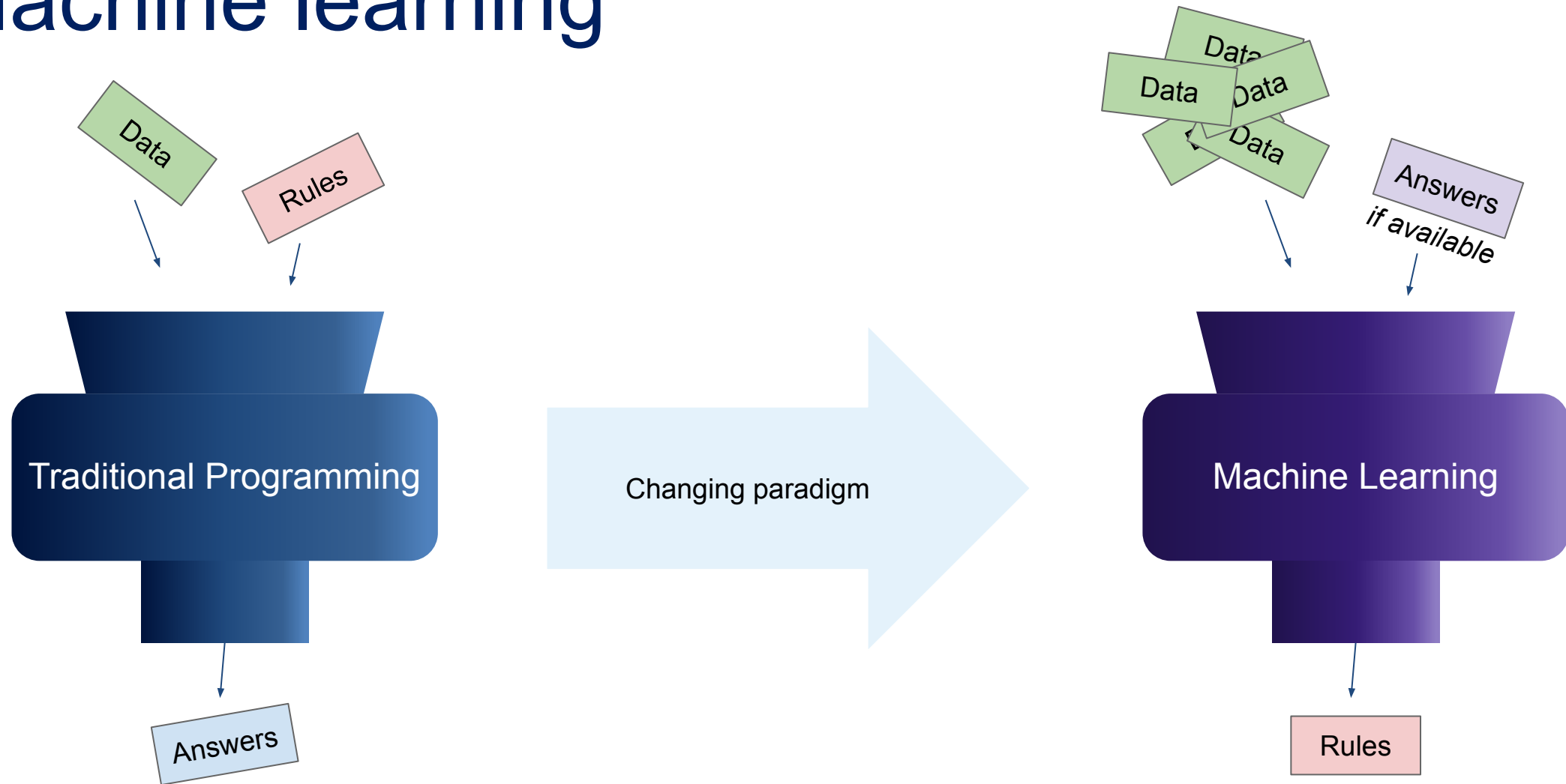


Learning $\neq$ Intelligence

- **Learning:** the capacity to ***change*** future behavior based on experience to improve some performance.
- **Intelligence:** the capacity to ***choose*** effective actions, solve problems, reason, or pursue goals across situations.



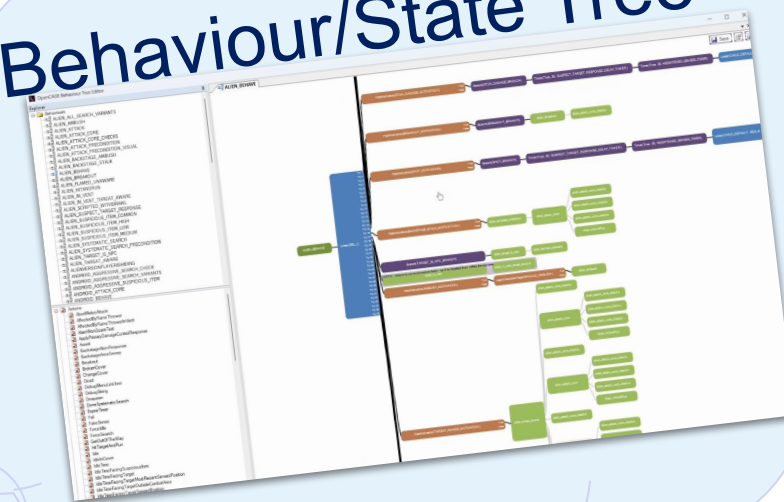
Machine learning



Artificial Intelligence
Can read inputs, reason
and make decisions.

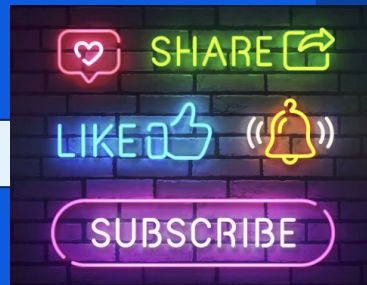


Behaviour/State Tree

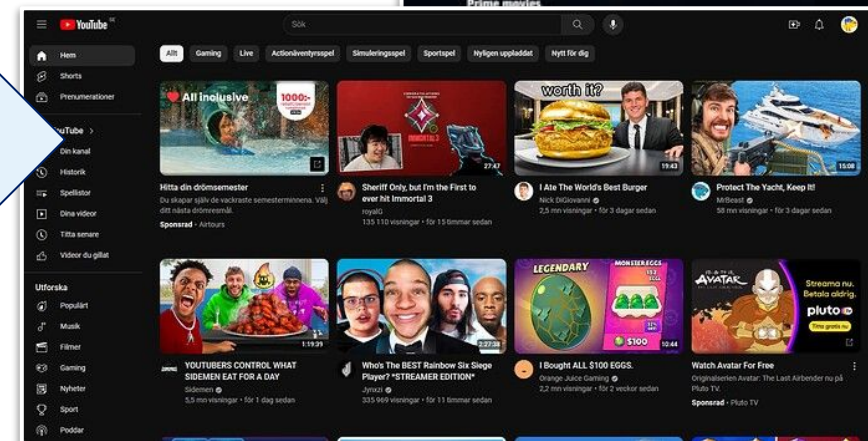


Artificial Intelligence
Can read inputs, reason
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Machine Learning
Can process feedback to the
decisions it took and change
its behaviour.



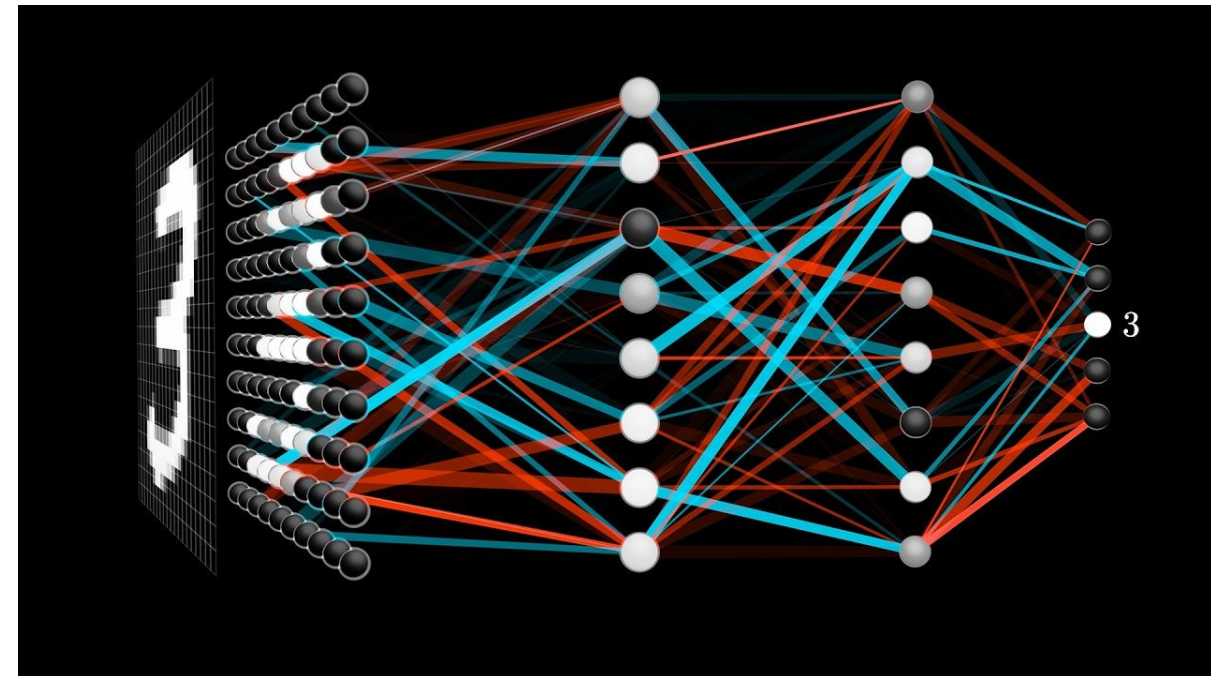
Recommendation algorithms



Artificial Intelligence
Can read inputs, reason
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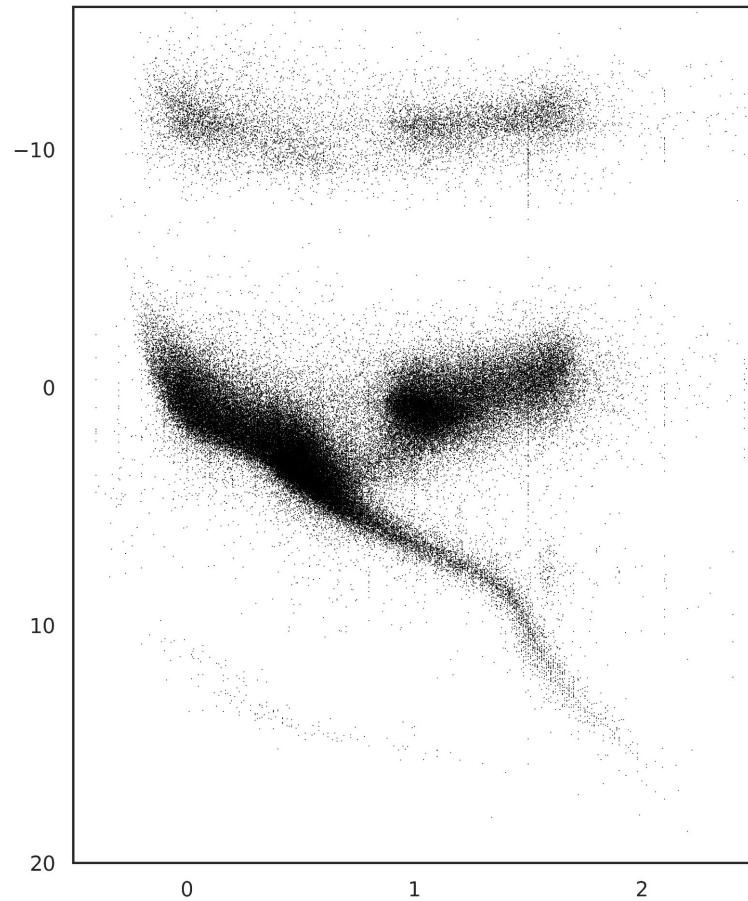
Deep Learning
“Deep” Artificial
Neural Networks



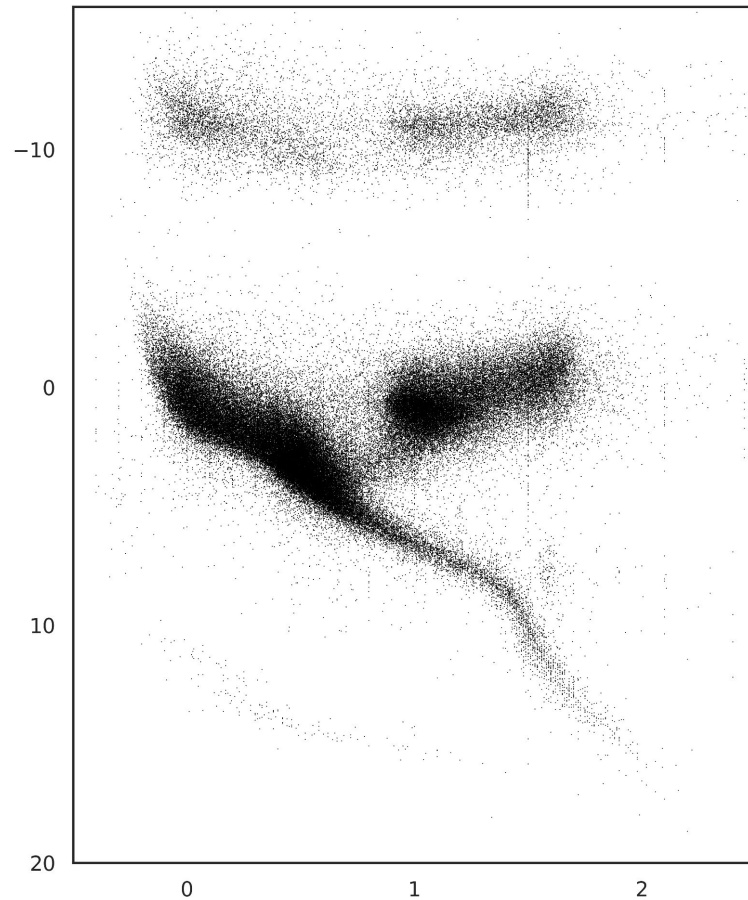
3Blue1Brown ✓
8,55 mi de inscritos

Deep learning chapter 1
<https://youtu.be/aircAruvnKk>

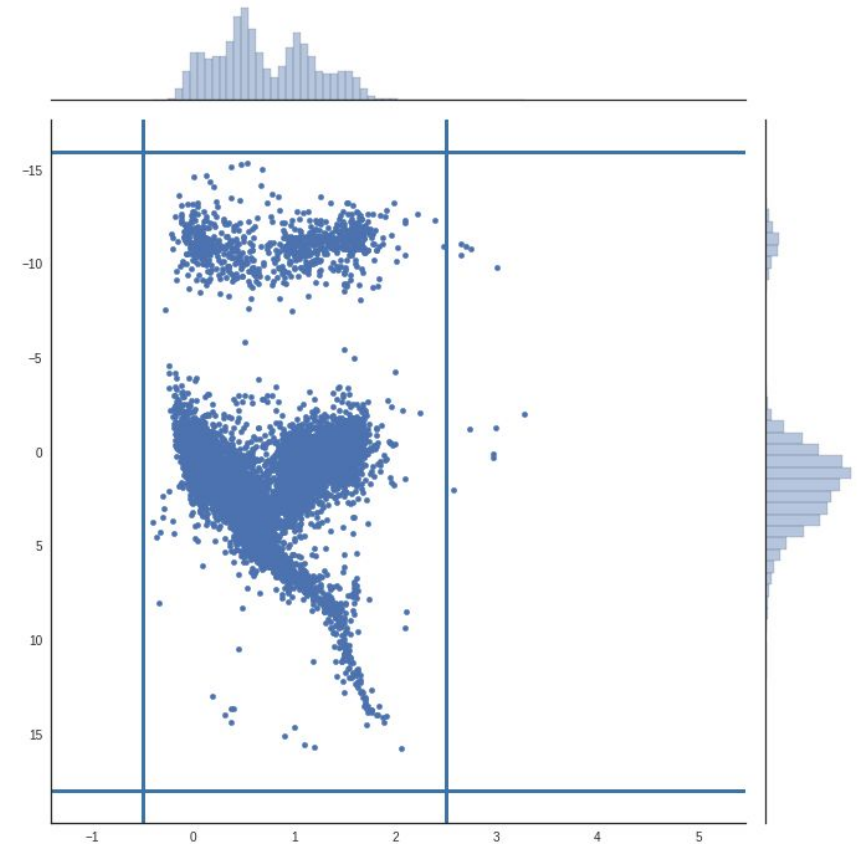
What can we learn from data?



What can we learn from data?

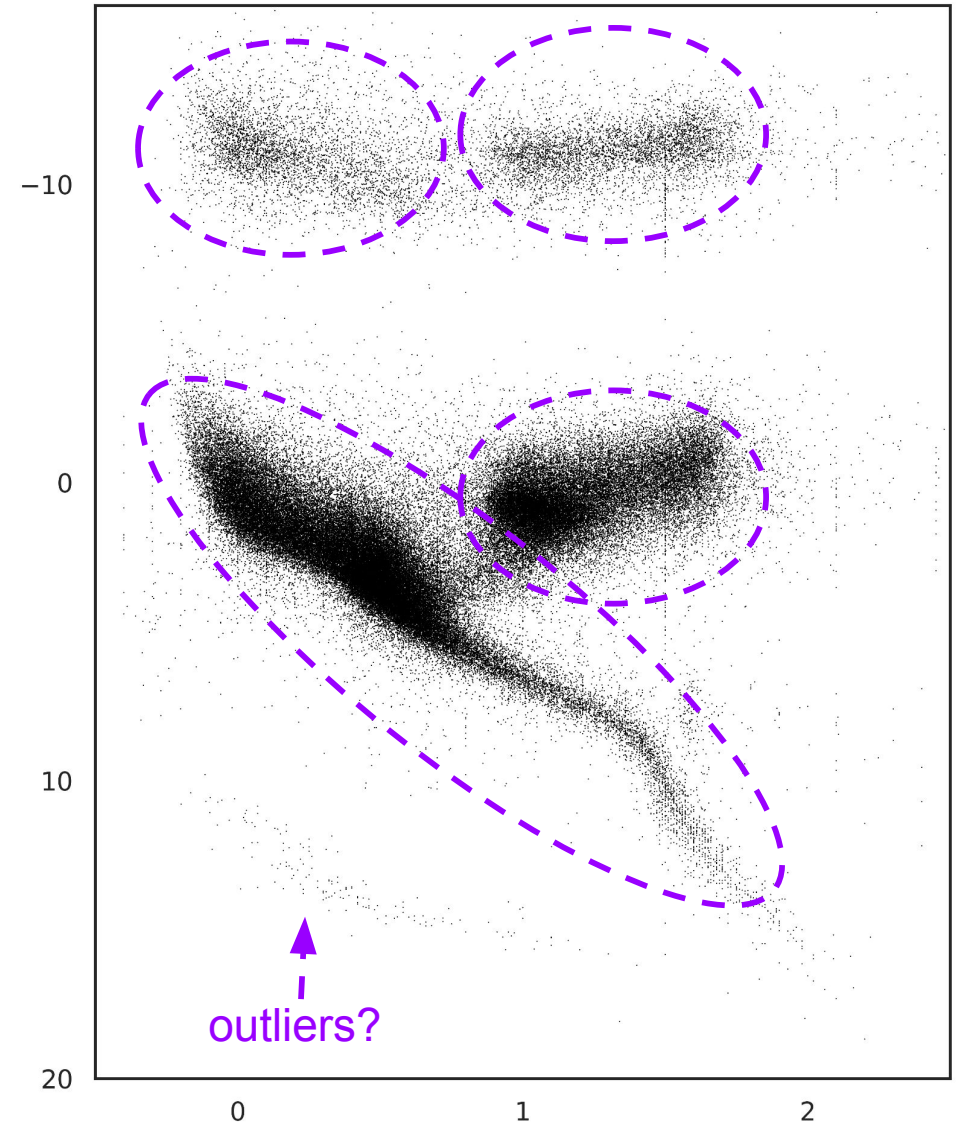


Statistical analysis



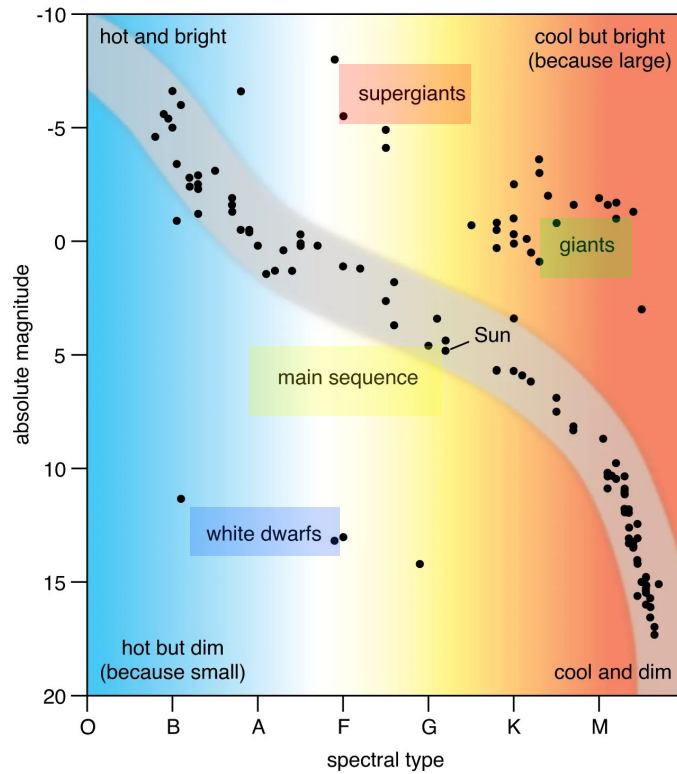
What can we learn from data?

Clustering
*Trying to group data
by some kind of
similarity.*



What can we learn from data?

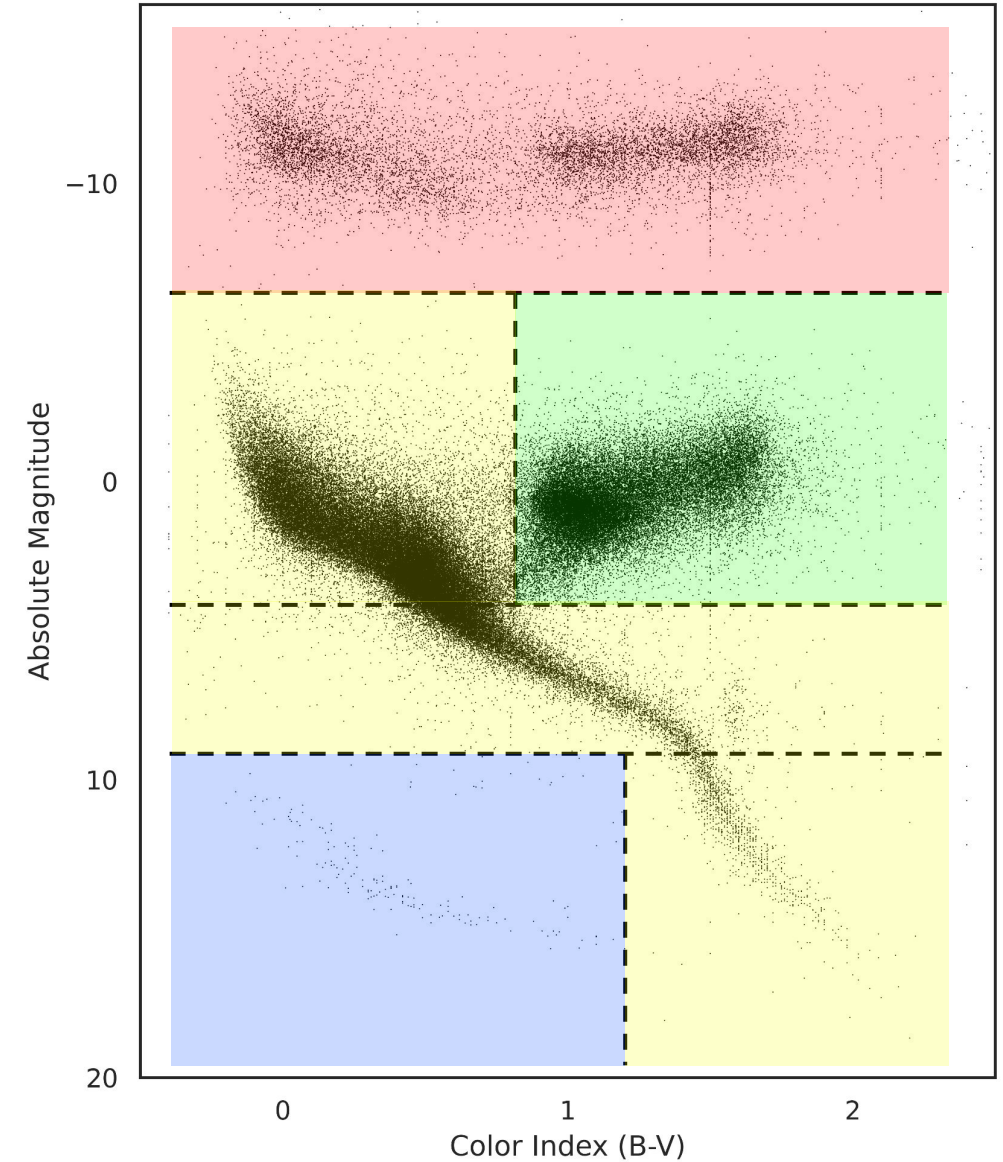
Hertzsprung-Russell diagram



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Classification
*Learn to classify
data based on
examples.*

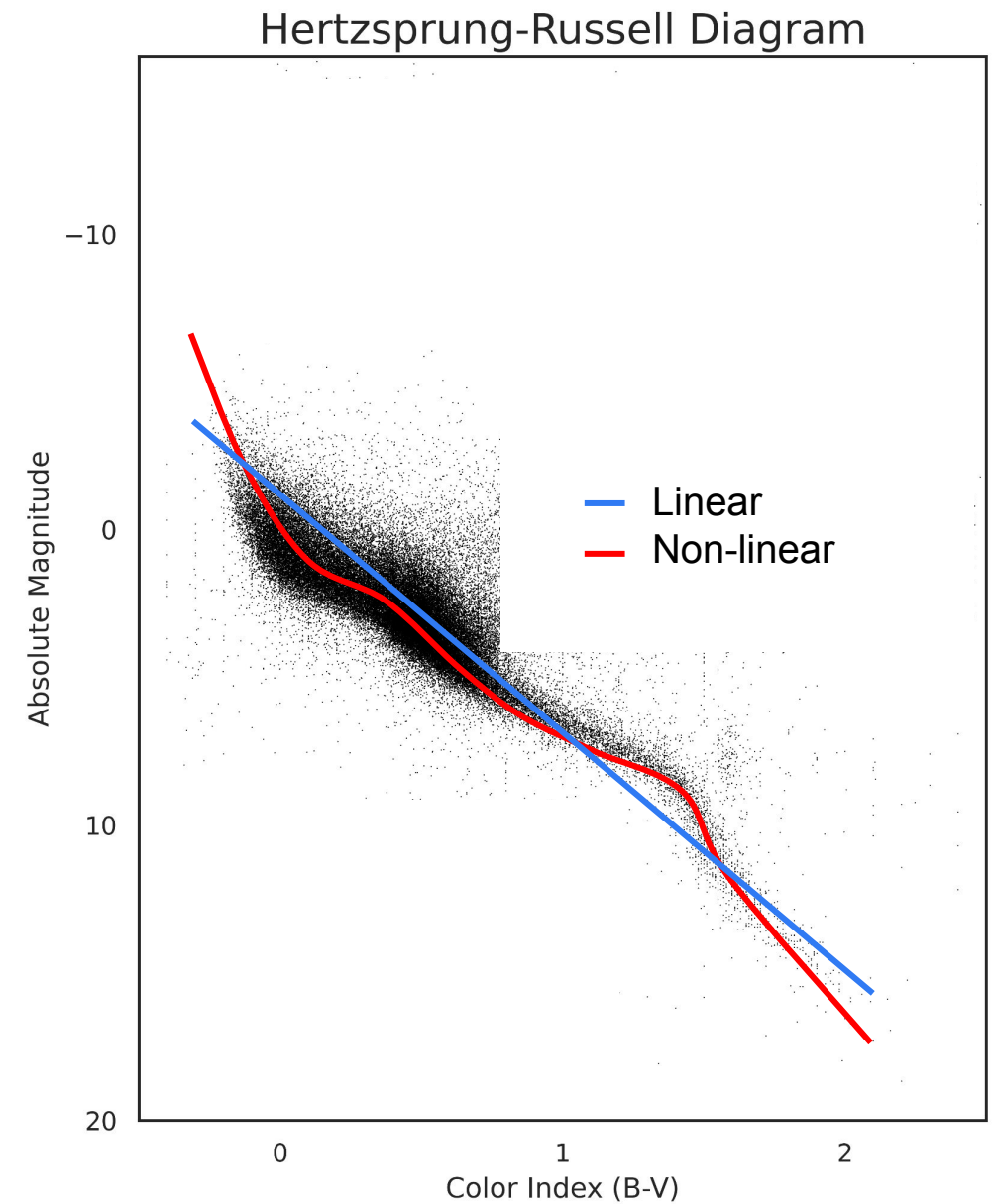
Hertzsprung-Russell Diagram



What can we learn from data?

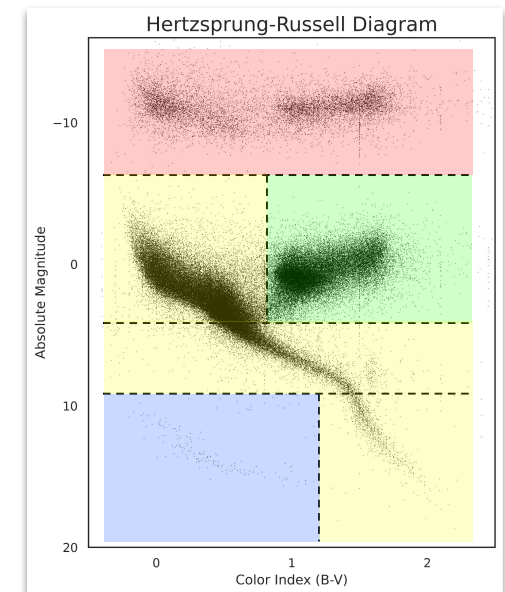
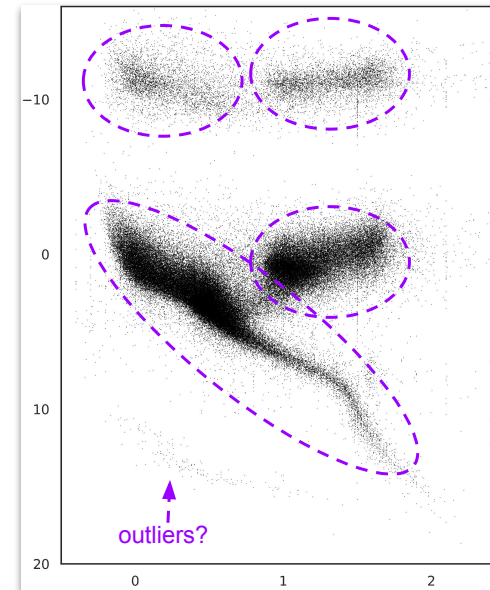
Regression

Learn the relation between quantities in our data based on examples.



Some more vocabulary:

- When we extract patterns from the data alone, this is called **Unsupervised** learning.
- When patterns are learned from examples, we call this learning **Supervised** learning
- When data comes with a assigned value to be learned, this data is called **labelled**.

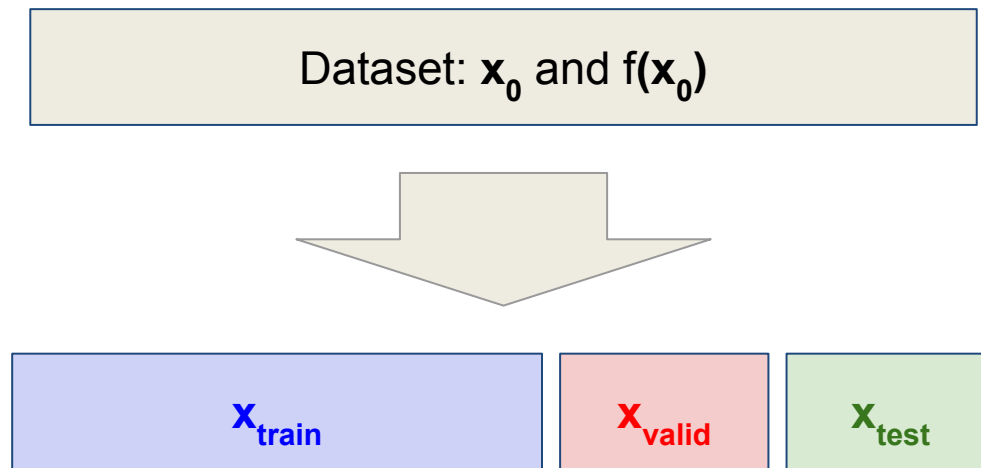


IA is not really intelligent

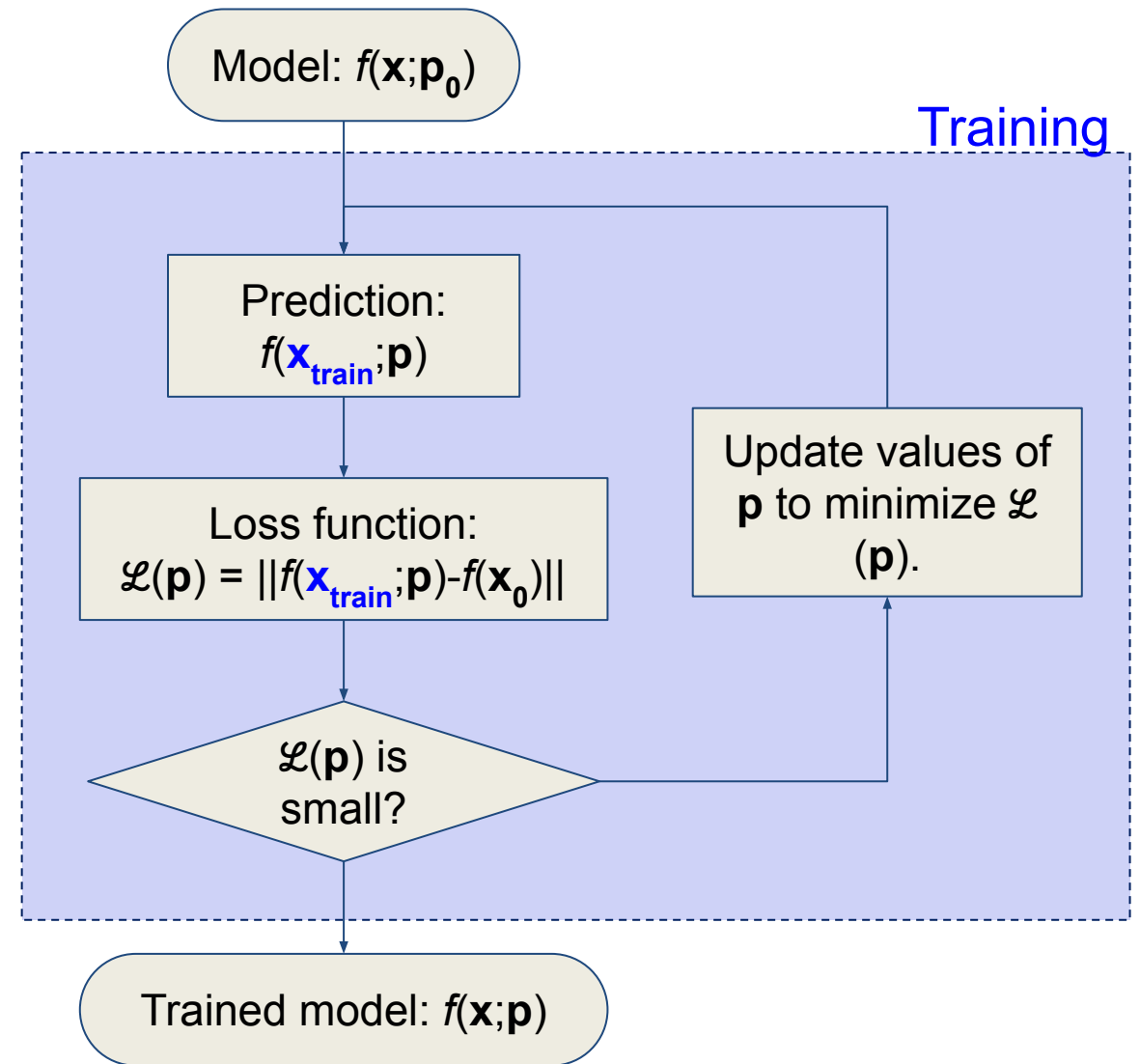
- The process of learning from learning from the data is called **training**. It is tricky to do it right...
- In the case of supervised learning, we typically split the dataset in tree parts: **training set**, **validation set**, and **test set**.



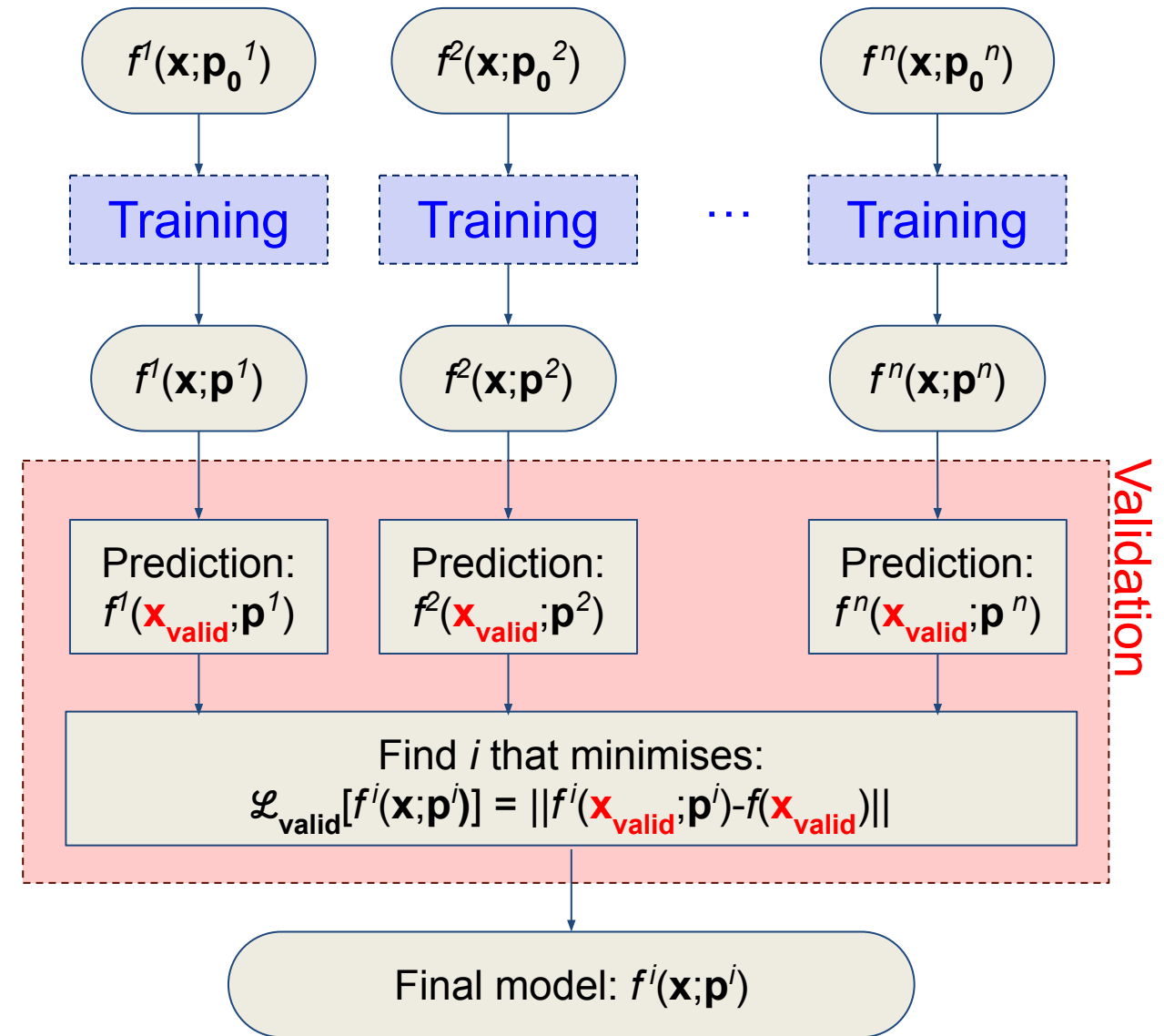
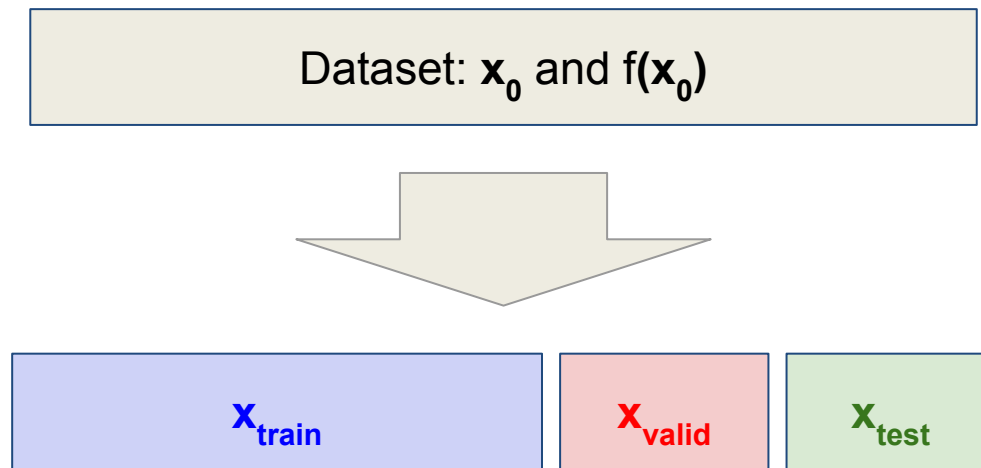
How learning works:



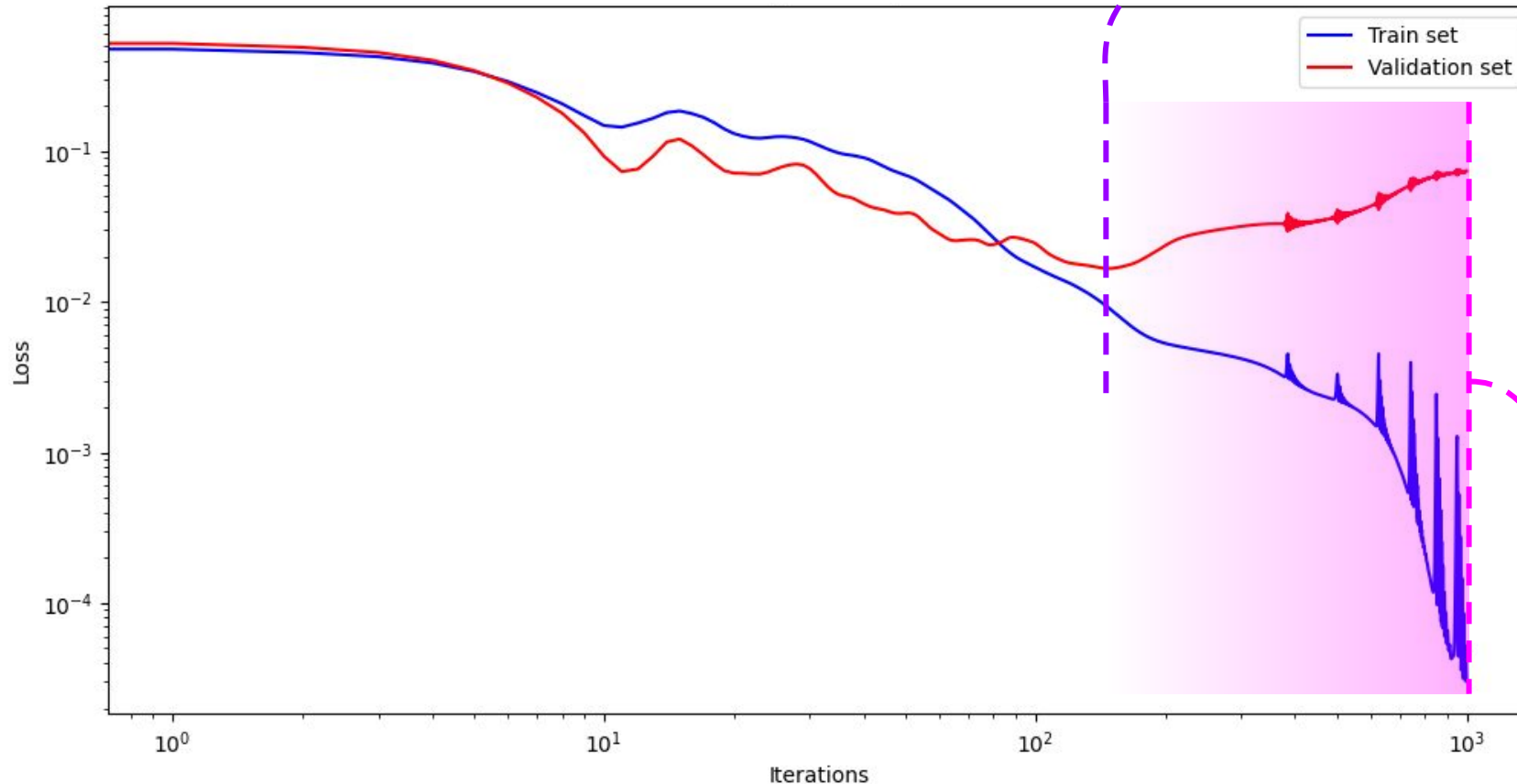
- Training is the optimization of a function called Loss ($\mathcal{L}$), that says how bad/good your model is in the training data.



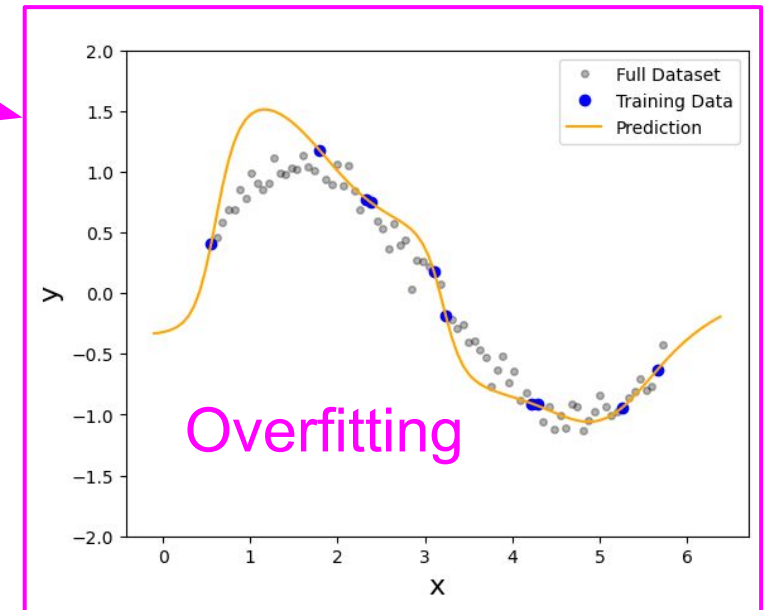
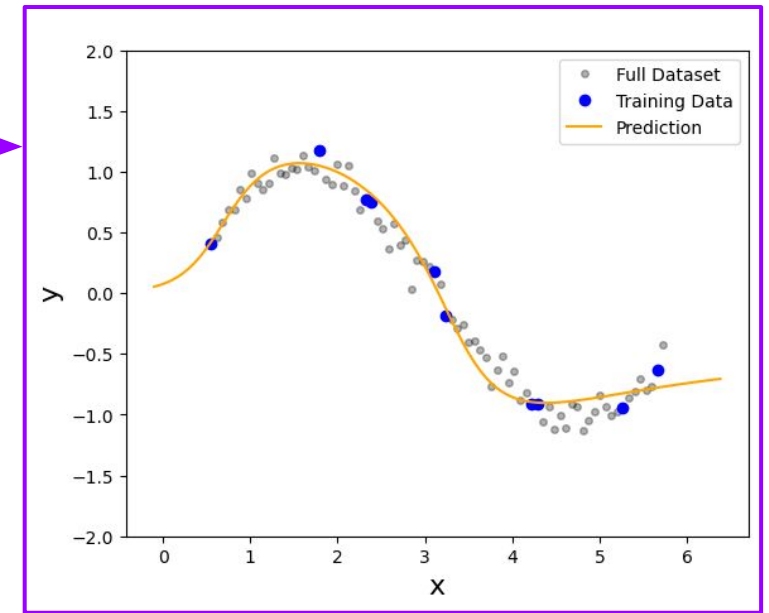
How learning works:



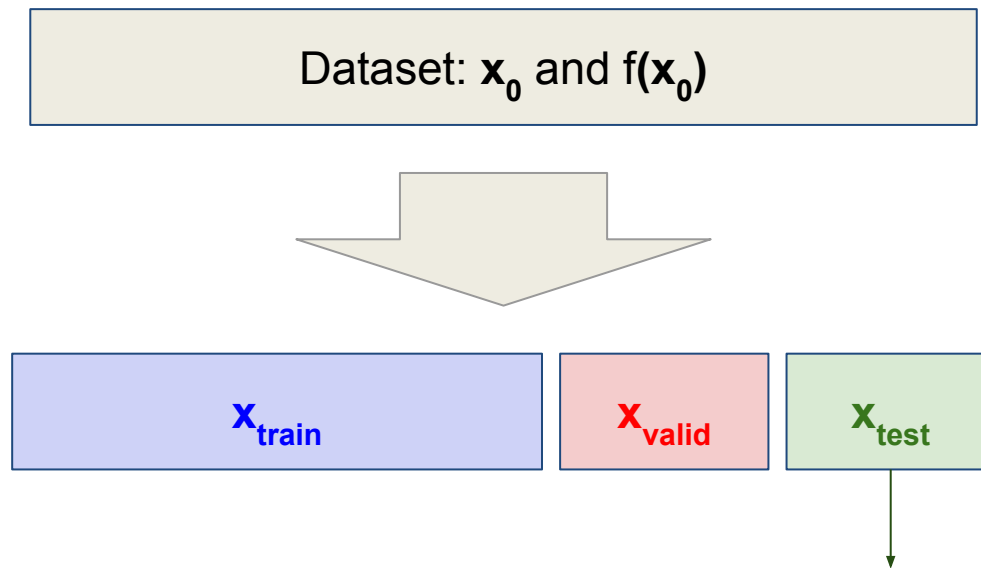
Early stopping:



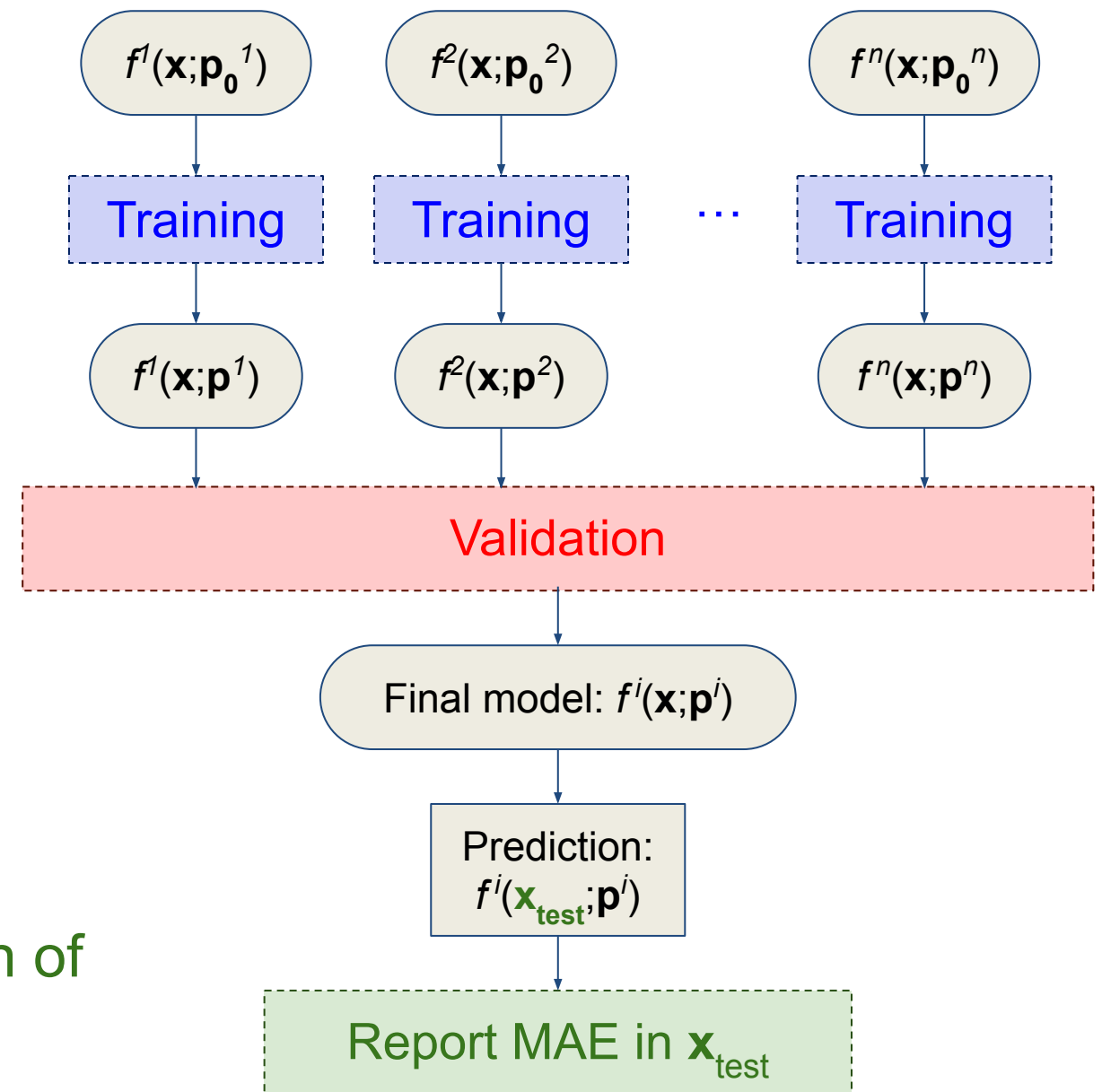
- Very likely if more degrees of freedom in the model than data to train.



And how good is your model?



- Only data never used in the obtention of the model can be used to test model quality or generality.



scikit-learn

Machine Learning in Python

[Getting Started](#)
[Release Highlights for 1.9](#)

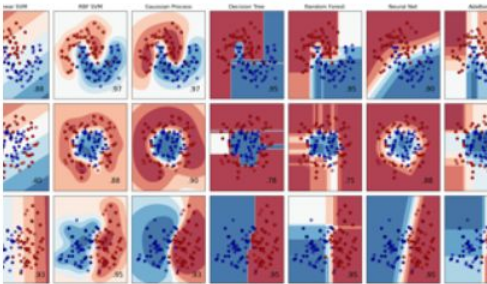
- Simple and efficient tools for predictive data analysis
- Accessible to everybody, and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib
- Open source, commercially usable - BSD license

Classification

Identifying which category an object belongs to.

Applications: Spam detection, image recognition.

Algorithms: [Gradient boosting](#), [nearest neighbors](#), [random forest](#), [logistic regression](#), and [more...](#)



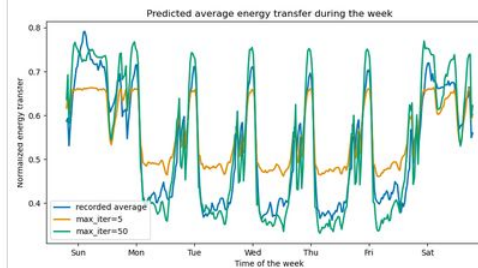
Examples

Regression

Predicting a continuous-valued attribute associated with an object.

Applications: Drug response, stock prices.

Algorithms: [Gradient boosting](#), [nearest neighbors](#), [random forest](#), [ridge](#), and [more...](#)



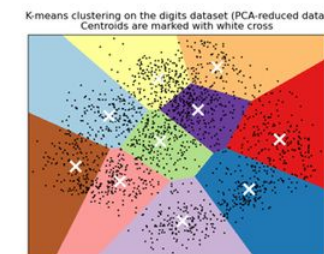
Examples

Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, grouping experiment outcomes.

Algorithms: [k-Means](#), [HDBSCAN](#), [hierarchical clustering](#), and [more...](#)



Examples

Dimensionality reduction

Reducing the number of random variables to consider.

Applications: Visualization, increased efficiency.

Algorithms: [PCA](#), [feature selection](#), [non-negative matrix factorization](#), and [more...](#)

Model selection

Comparing, validating and choosing parameters and models.

Applications: Improved accuracy via parameter tuning.

Algorithms: [Grid search](#), [cross validation](#), [metrics](#), and [more...](#)

Preprocessing

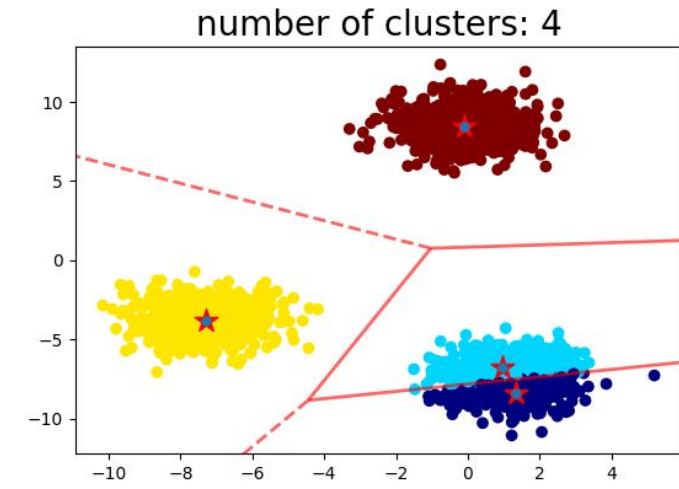
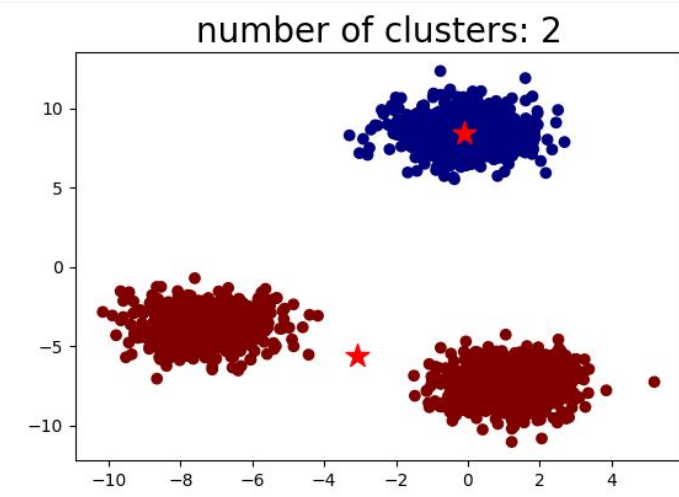
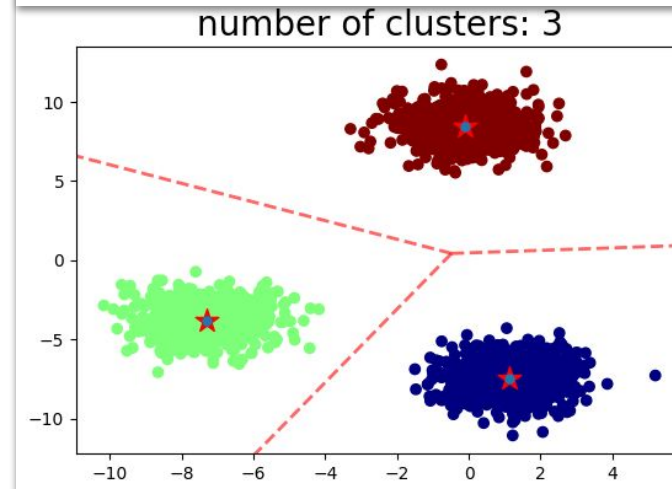
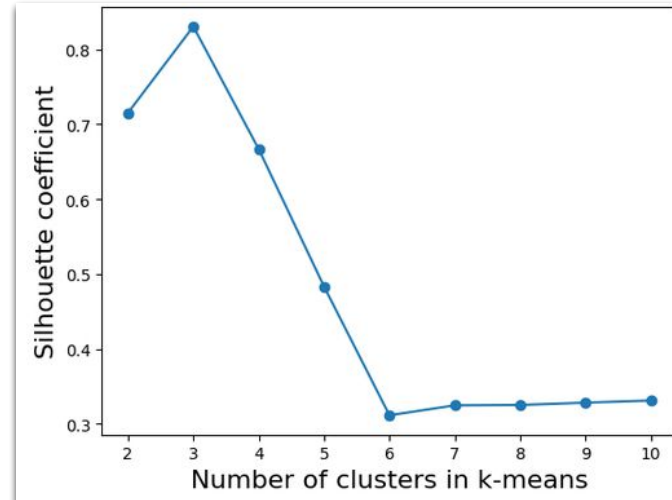
Feature extraction and normalization.

Applications: Transforming input data such as text for use with machine learning algorithms.

Algorithms: [Preprocessing](#), [feature extraction](#), and [more...](#)

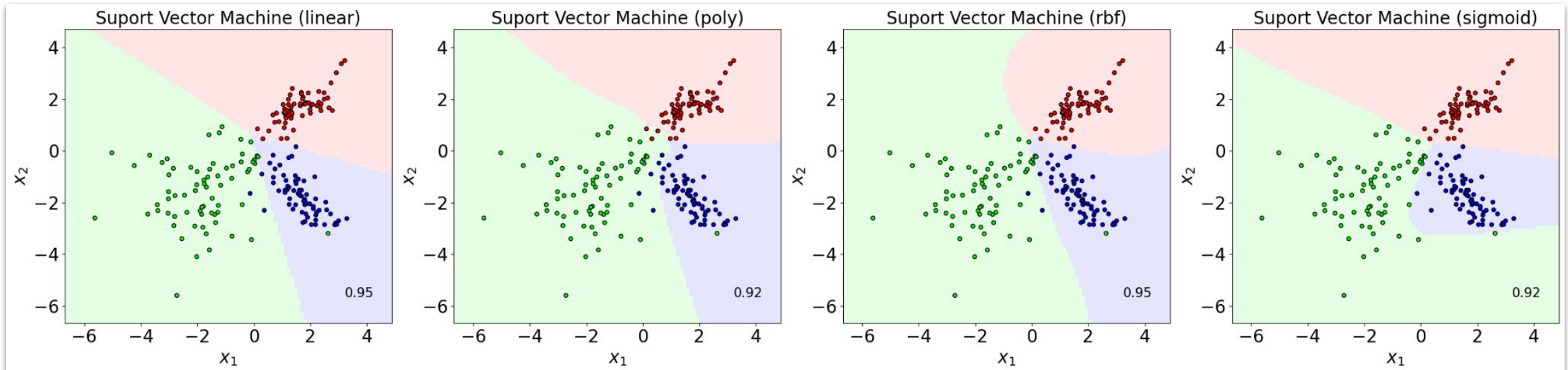
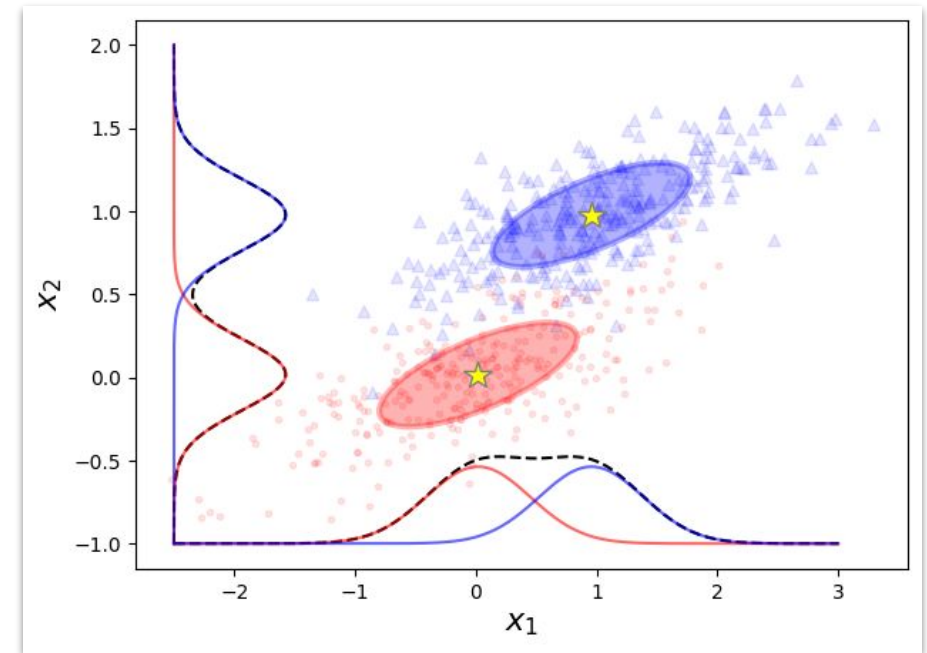
In the hands-on:

- 01 - Clustering
 - K-means
 - Gaussian Mixture Model
 - Hierarchical Clustering
 - DBSCAN



In the hands-on section:

- 02 - Classification
 - Linear Discriminant Analysis
 - Support Vector Machine
 - k-Nearest Neighbors

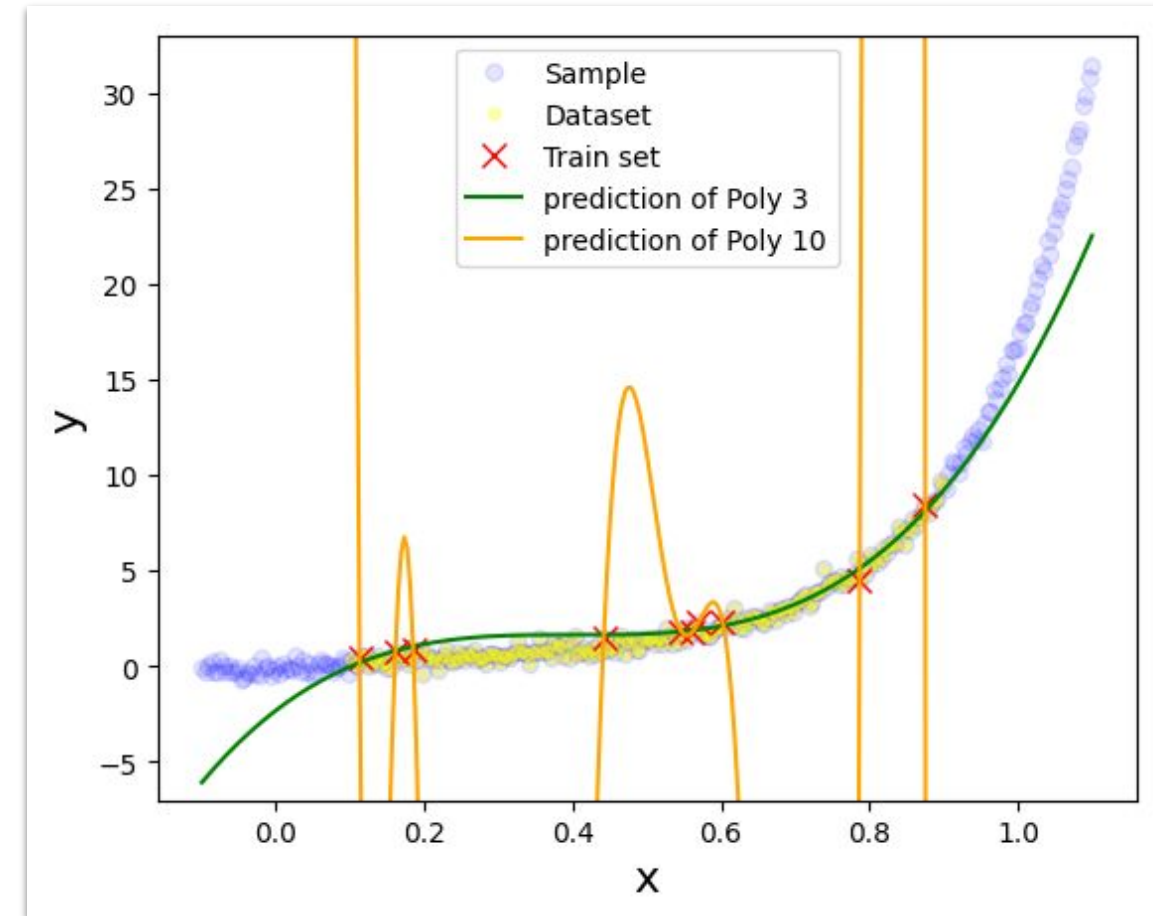


In the hands-on section:

- 03 - Regression
 - Linear Regression
 - Polynomial Regression

⚠ Artificial Neural Network ⚠

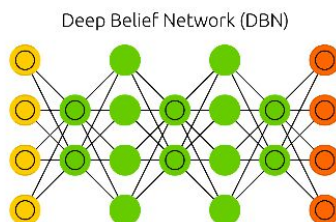
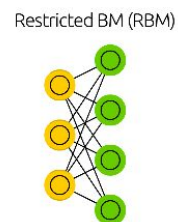
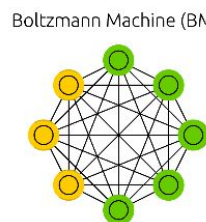
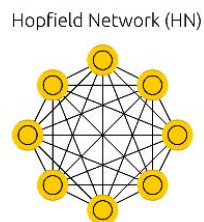
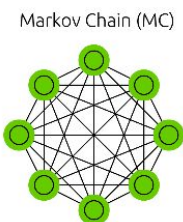
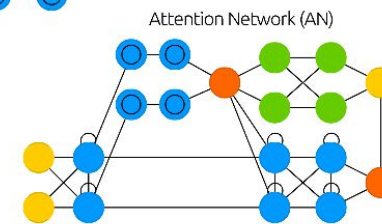
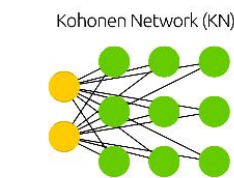
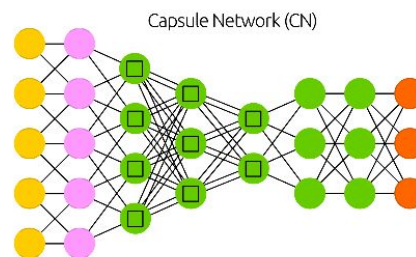
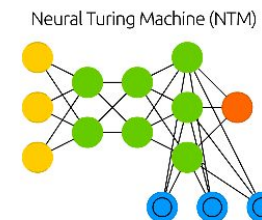
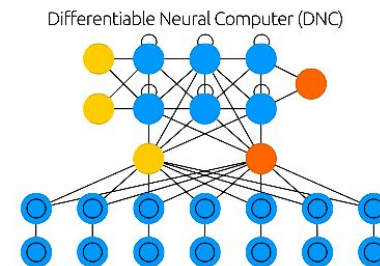
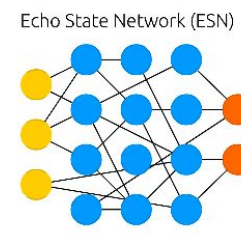
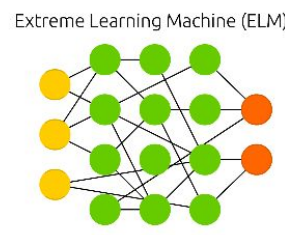
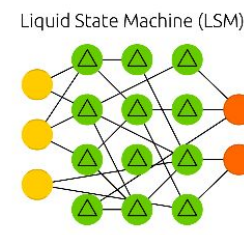
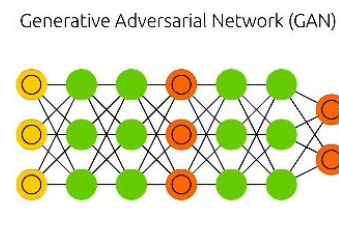
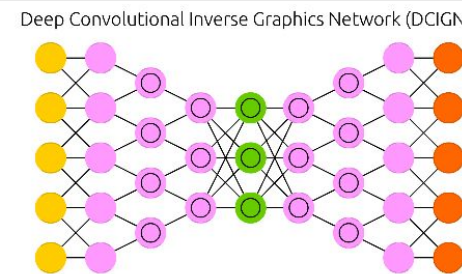
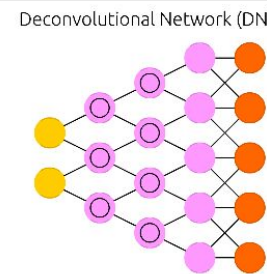
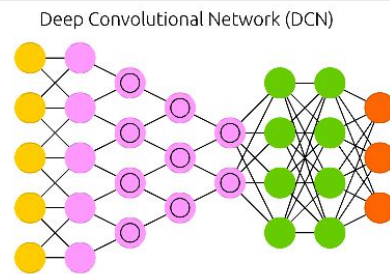
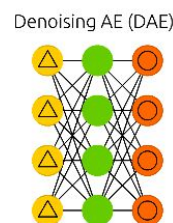
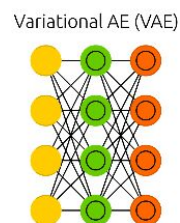
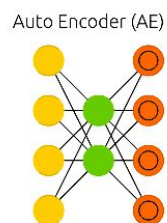
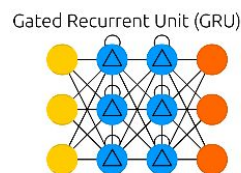
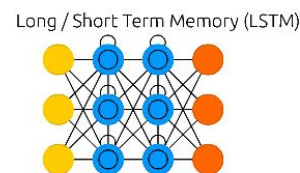
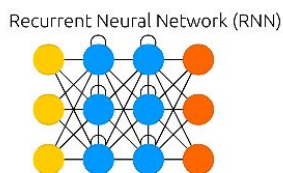
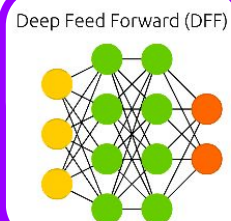
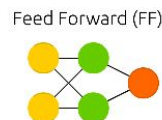
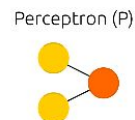
- Multilayer Perceptron from sklearn
- Multilayer Perceptron from PyTorch



A mostly complete chart of Neural Networks

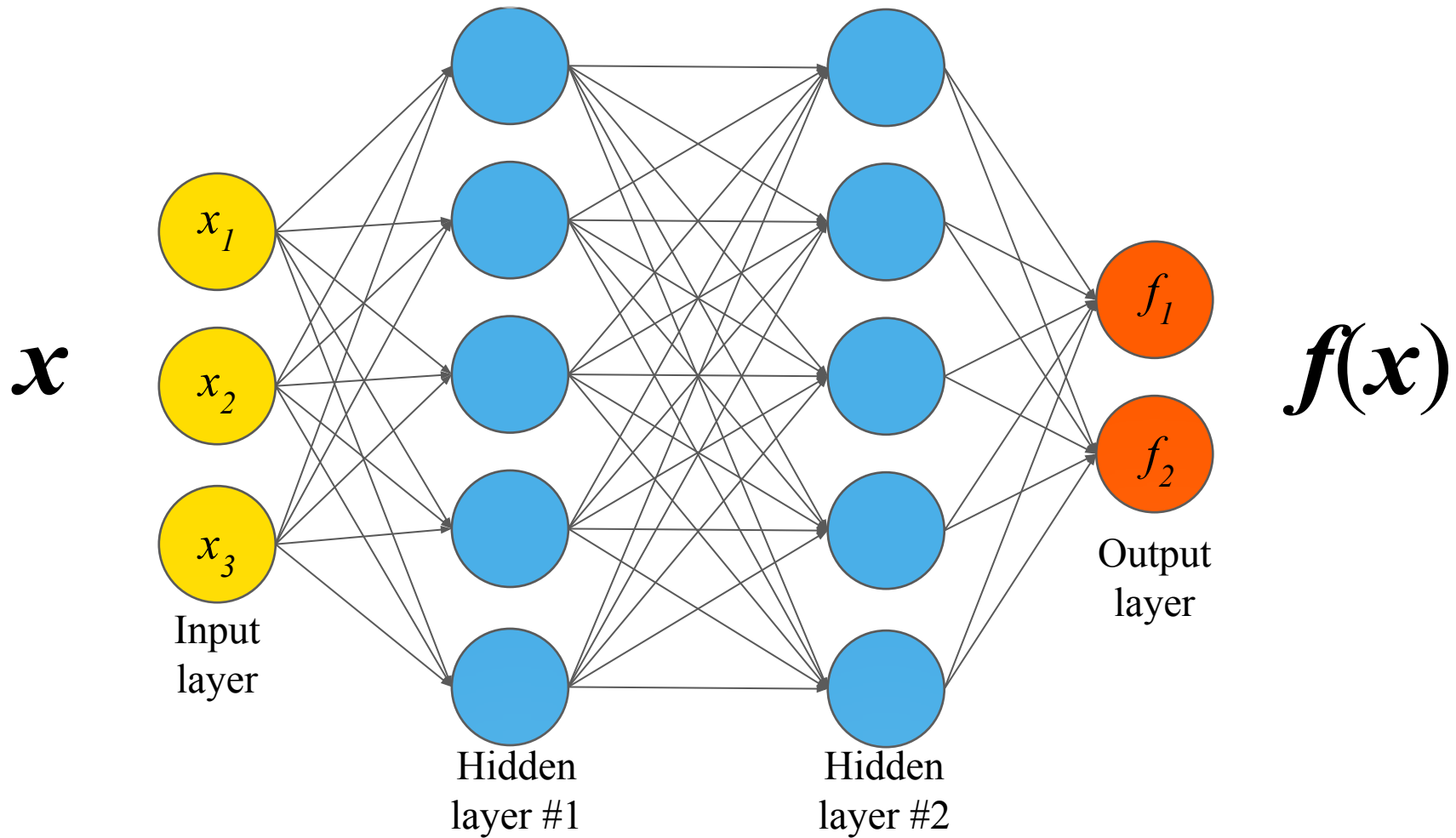
©2019 Fjodor van Veen & Stefan Leijnen asimovinstitute.org

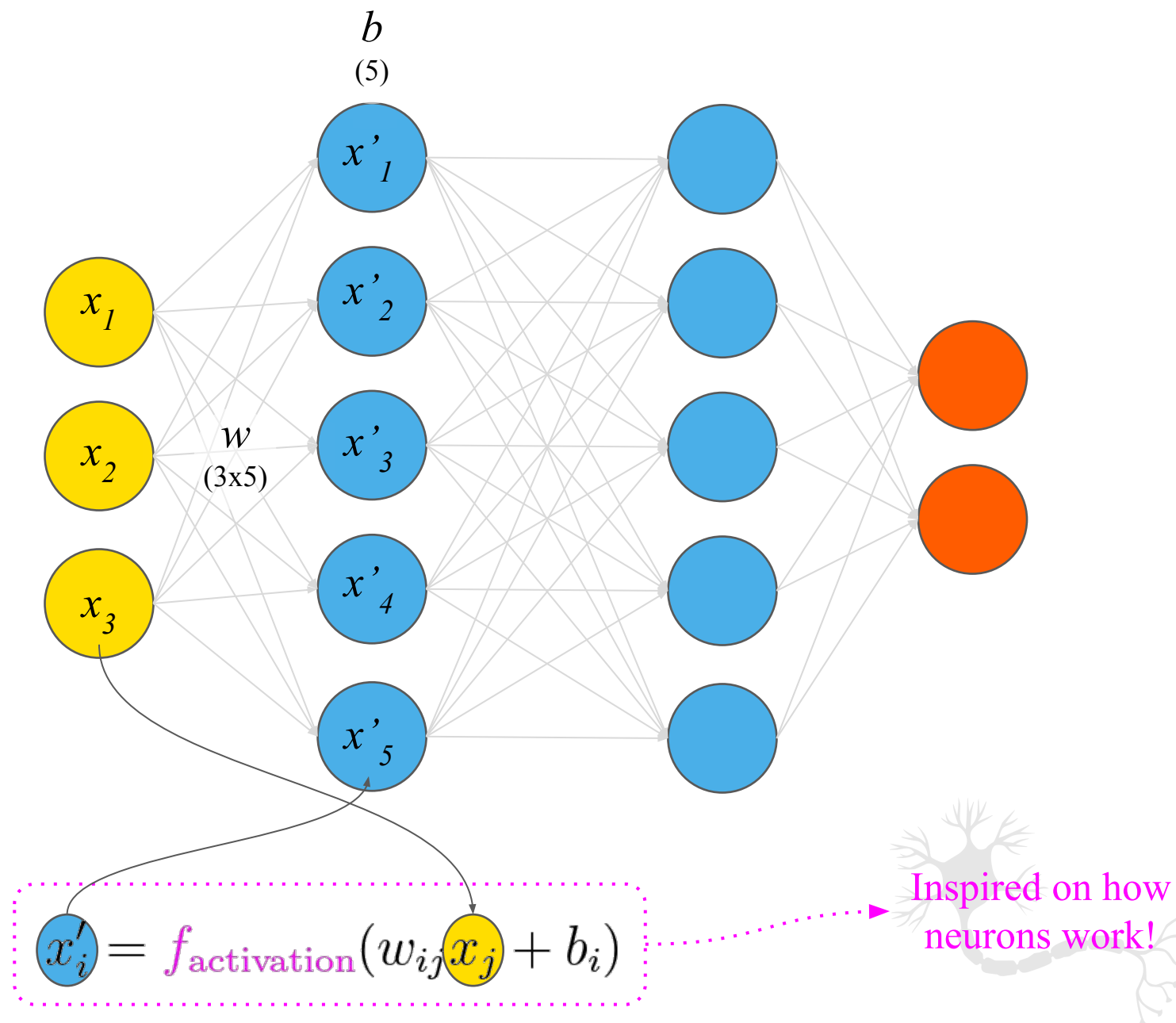
-  Input Cell
-  Backfed Input Cell
-  Noisy Input Cell
-  Hidden Cell
-  Probabilistic Hidden Cell
-  Spiking Hidden Cell
-  Capsule Cell
-  Output Cell
-  Match Input Output Cell
-  Recurrent Cell
-  Memory Cell
-  Gated Memory Cell
-  Kernel
-  Convolution or Pool



The Neural Network Zoo (2020) - The Asimov Institute

Feed Forward Neural Network





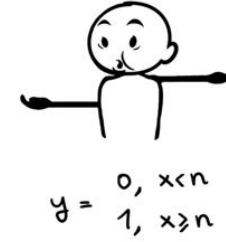
Sigmoid



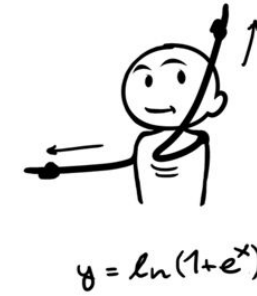
Tanh



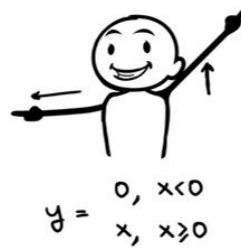
Step Function



Softplus



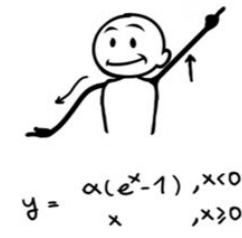
ReLU



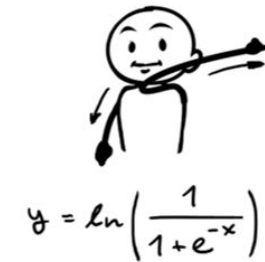
Softsign



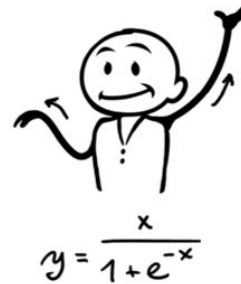
ELU



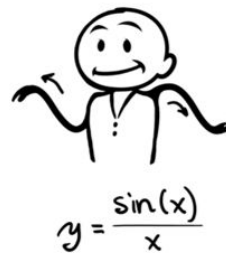
Log of Sigmoid



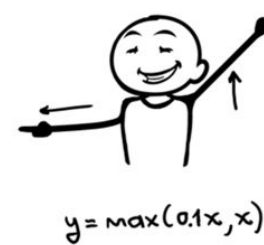
Swish



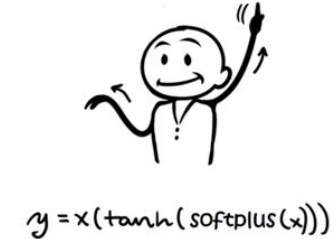
Sinc



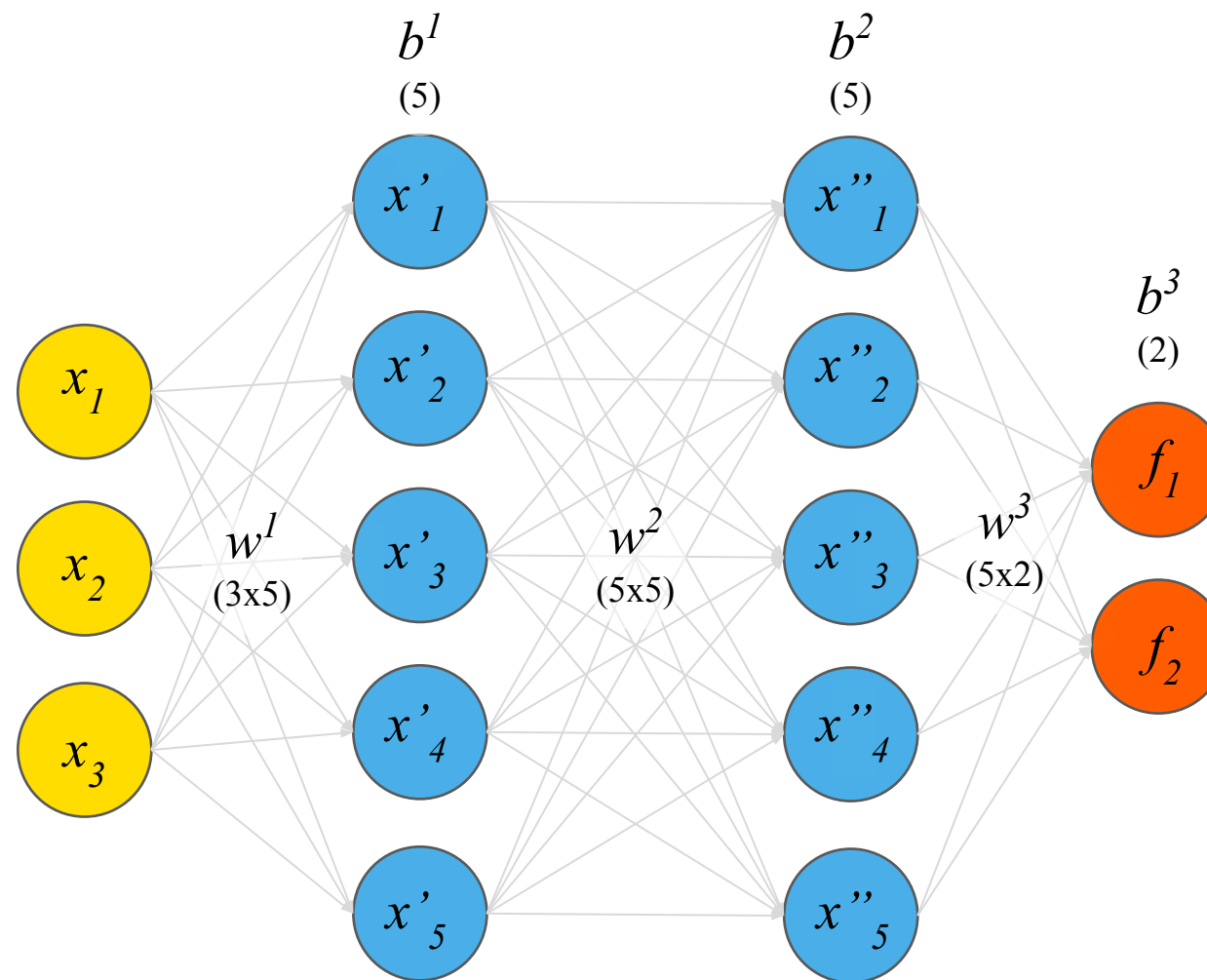
Leaky ReLU



Mish



Illustrations by Sefik Ilkin Serengil

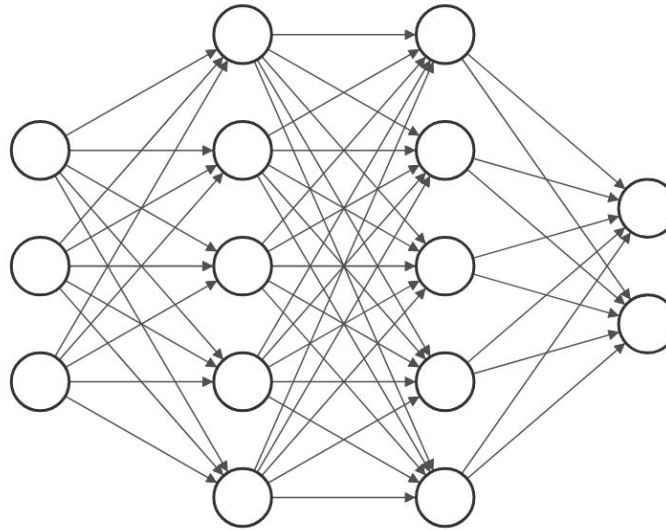


$$x'_i = f_{\text{activation}}(w_{ij}^1 x_j + b_i^1)$$

$$f_i = f_{\text{activation}}(w_{ij}^3 x'_j + b_i^3)$$

$$x''_i = f_{\text{activation}}(w_{ij}^2 x'_j + b_i^2)$$

We draw this



just so we don't scare people with

$$\text{out}(\mathbf{x}) = f(\mathbf{W}_3 \times f(\mathbf{W}_2 \times f(\mathbf{W}_1 \times \text{inp} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3)$$

*At the end, ANN are just **functions** that we represent with graphs and parametrize with training data.*

Why do regression with ANNs?

- Universal Approximation Theorems (UATs): Neural Networks are shown to be good **universal approximators** even in the case of intricate functions.
- There are versions of those theorems, including large enough width and/or large enough depth.

Multilayer Feedforward Networks are Universal Approximators

KUR' HORNIK

Technische Universität Wien

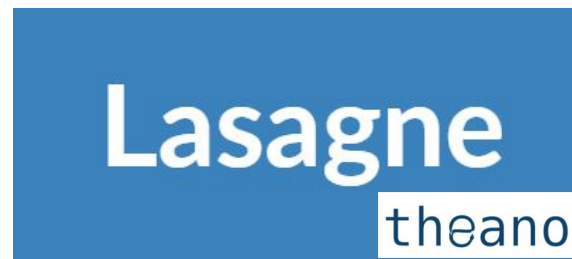
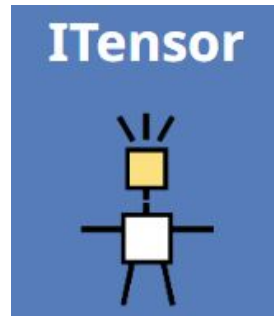
MAXWELL STINCHCOMBE AND HALBER WHITE

University of California, San Diego

(Received 16 September 1988; revised and accepted 9 March 1989)

Abstract—This paper rigorously establishes that standard multilayer feedforward networks with as few as one hidden layer using arbitrary squashing functions are capable of approximating any Borel measurable function from one finite dimensional space to another to any desired degree of accuracy, provided sufficiently many hidden units are available. In this sense, **multilayer feedforward networks are a class of universal approximators.**

Some tools and libraries for ANNs



Caffe





PROGRAMME
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